

Boundary Design Patterns

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Controlling Data Visibility with Boundaries

Boundaries determine **what data** users can see. While policies control actions, boundaries filter visibility. This notebook covers boundary syntax, patterns, and implementation strategies.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM enabled
Permissions	<code>environment-admin</code> or boundary management rights
Prior Knowledge	IAMADM-01 through IAMADM-04

1. Boundary Fundamentals

Boundaries filter what entities and data a user can see within an environment.

Policies vs Boundaries

Concept	Controls	Example
Policy	What actions	"Can read logs"
Boundary	What data	"Logs from checkout service only"

A user needs BOTH:

- A **policy** granting the action (e.g., `storage:logs:read`)
- A **boundary** including the data (e.g., `checkout` security context)

How Boundaries Work

1. User attempts to access data (query, view, etc.)
2. Dynatrace checks if user's policy allows the action
3. Dynatrace filters results to match user's boundary
4. User sees only data within their boundary

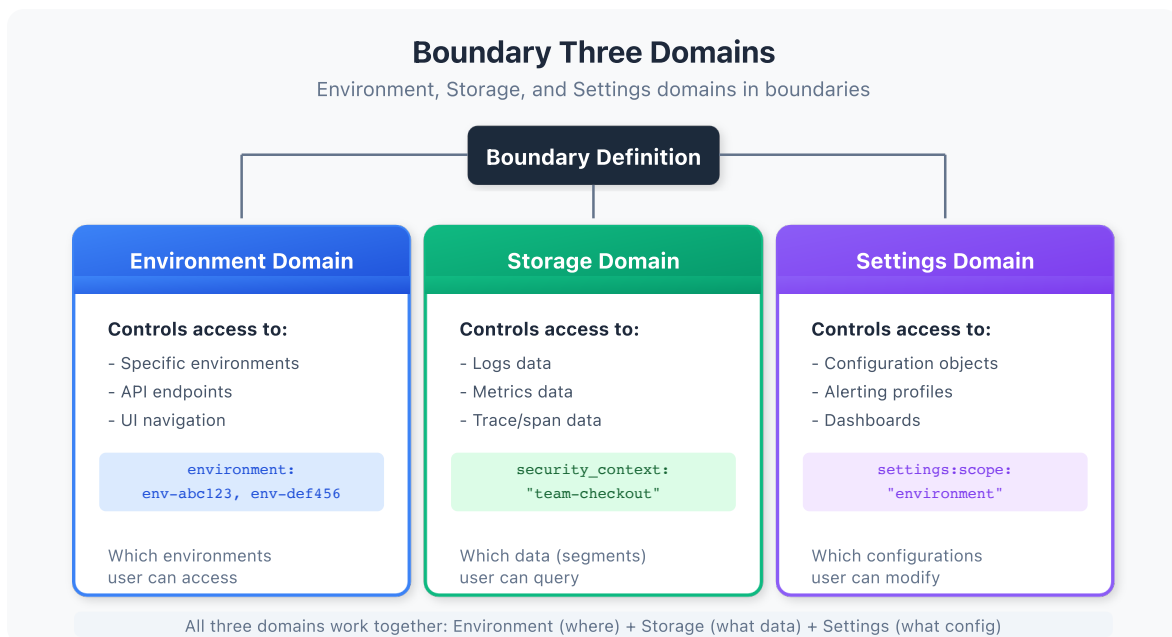
Boundary Scope

Boundaries are **environment-level** and assigned to groups:

```
Group: dt-checkout-editors
├─ Policy: environment-editor
└─ Boundary: checkout-services-only
```

2. The Three-Domain Model

Boundaries filter across three domains. A complete boundary typically includes all three.



Domain 1: Environment

Controls visibility of Smartscape entities:

- Hosts
- Services
- Applications
- Databases
- Cloud resources

Domain 2: Storage

Controls visibility of Grail data:

- Logs
- Spans (traces)

- Metrics
- Events
- Business events

Domain 3: Settings

Controls visibility of configuration:

- Alerting rules
- SLOs
- Custom settings

Why All Three Matter

If you only boundary one domain:

Missing Domain	Problem
Environment	User can't see services in UI
Storage	User can't query logs/spans
Settings	User can't see/edit config

3. Boundary Syntax Reference

Basic Boundary Structure

```
<domain>:<field> <operator> (<values>)
```

Complete Three-Domain Boundary:

```
environment:dt.security_context IN ("checkout");  
storage:dt.security_context IN ("checkout");  
settings:dt.security_context IN ("checkout");
```

Operators

Operator	Description	Example
<code>IN</code>	Match any in list	<code>IN ("a", "b", "c")</code>
<code>=</code>	Exact match	<code>= "checkout"</code>
<code>!=</code>	Not equal	<code>!= "restricted"</code>
<code>startsWith</code>	Prefix match	<code>startsWith "team-"</code>
<code>contains</code>	Substring	<code>contains "prod"</code>

Common Fields

Domain	Field	Description
environment	<code>dt.security_context</code>	Entity security context
environment	<code>management-zone</code>	Legacy MZ (deprecated)
storage	<code>dt.security_context</code>	Data security context
storage	<code>bucket.name</code>	Storage bucket name
settings	<code>dt.security_context</code>	Settings security context
settings	<code>schemaId</code>	Settings schema

Multiple Values

Grant access to multiple security contexts:

```
environment:dt.security_context IN ("checkout", "payments", "shared");
storage:dt.security_context IN ("checkout", "payments", "shared");
settings:dt.security_context IN ("checkout", "payments", "shared");
```

Wildcard Access

For admin groups needing full access:

```
environment:dt.security_context IN ("*");  
storage:dt.security_context IN ("*");  
settings:dt.security_context IN ("*");
```

4. Security Context Strategy

Security contexts are the foundation of boundary filtering. Plan your strategy carefully.

What is a Security Context?

A **security context** is a label applied to:

- Entities (hosts, services)
- Data (logs, spans)
- Settings (configurations)

Boundaries filter based on these labels.

Assigning Security Context

Security context is set via (modern approaches):

1. **Entity Enrichment rules** - Automatic based on entity properties
2. **Host properties** - Set on hosts and inherited by services
3. **Primary Grail tags** - First-class tags for Grail data
4. **OneAgent group** - Inherited from deployment
5. **Log attributes** - Set during OpenPipeline ingestion

Note: Auto-tagging is a legacy approach. Prefer Entity Enrichment for new implementations.

Security Context Design Patterns

Pattern	Example Values	Use Case
Team-based	checkout-team , payments-team	Team ownership
Application-based	ecommerce-app , mobile-app	App isolation
Environment-based	prod , staging , dev	Env separation
Business unit	retail , wholesale , corporate	Business isolation
Geography	us-east , eu-west , apac	Regional separation

Naming Conventions

Rule	Good	Bad
Lowercase	checkout	Checkout
Hyphens	team-checkout	team_checkout
Descriptive	payments-api	pa
No spaces	mobile-app	mobile app

Shared Context

Some data should be visible to multiple teams:

```
Security Contexts:
├─ checkout      (checkout team data)
├─ payments      (payments team data)
└─ shared        (cross-team data - infrastructure, common services)
```

Teams get their context + shared :

```
environment:dt.security_context IN ("checkout", "shared");
```

5. Common Boundary Patterns

Pattern 1: Single Team Boundary

Restrict to one team's data:

```
environment:dt.security_context IN ("checkout");
storage:dt.security_context IN ("checkout");
settings:dt.security_context IN ("checkout");
```

Pattern 2: Team + Shared

Team data plus shared infrastructure:

```
environment:dt.security_context IN ("checkout", "shared", "infrastructure");
storage:dt.security_context IN ("checkout", "shared", "infrastructure");
settings:dt.security_context IN ("checkout", "shared");
```

Pattern 3: Multiple Teams (Cross-Functional)

For SRE or platform teams:

```
environment:dt.security_context IN ("checkout", "payments", "catalog",
"shared");
storage:dt.security_context IN ("checkout", "payments", "catalog", "shared");
settings:dt.security_context IN ("checkout", "payments", "catalog",
"shared");
```

Pattern 4: Environment Tier

Production vs non-production:

Production:

```
environment:dt.security_context IN ("prod-checkout", "prod-payments");
storage:dt.security_context IN ("prod-checkout", "prod-payments");
settings:dt.security_context IN ("prod-checkout", "prod-payments");
```

Non-Production:


```
environment:dt.security_context IN ("dev-checkout", "staging-checkout");
storage:dt.security_context IN ("dev-checkout", "staging-checkout");
settings:dt.security_context IN ("dev-checkout", "staging-checkout");
```

Pattern 5: Read-All, Write-Scoped

Using two groups for the same user:

Group 1: All-Viewers (broad read)

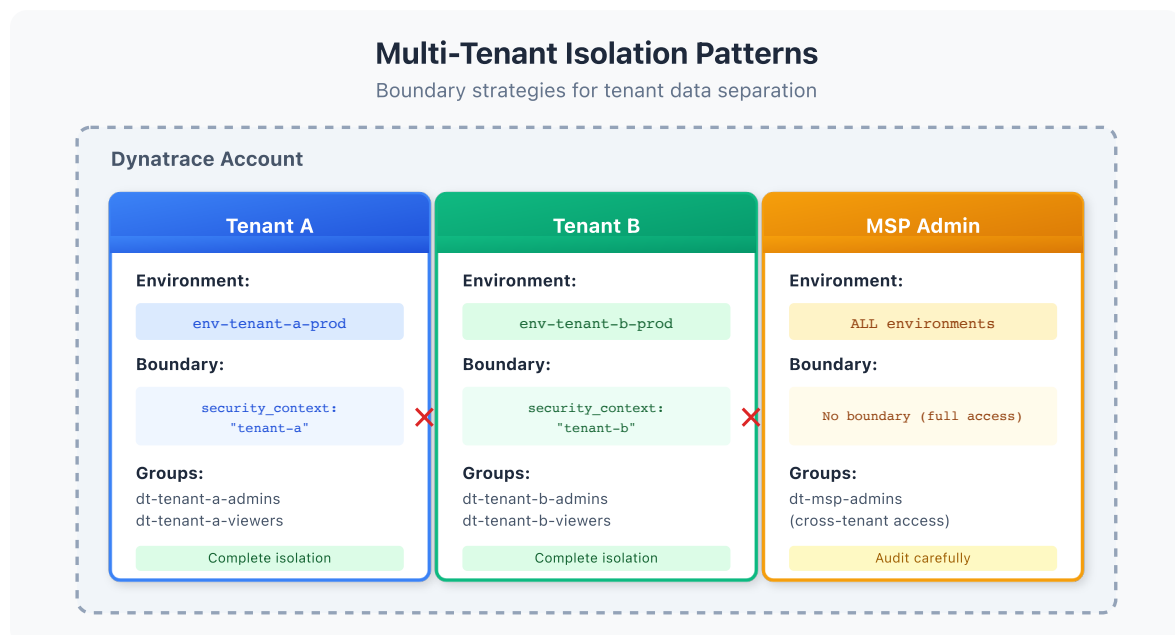
```
Policy: environment-viewer
Boundary: environment:dt.security_context IN ("*");
```

Group 2: Checkout-Editors (scoped write)

```
Policy: checkout-write-policy
Boundary: environment:dt.security_context IN ("checkout");
```

6. Multi-Tenant Isolation

For organizations serving multiple customers or business units that require strict isolation.



Isolation Requirements

Requirement	Implementation
Data isolation	Unique security context per tenant
No cross-tenant access	Strict boundary matching
Audit capability	Centralized logging
Shared infrastructure	Separate "platform" context

Multi-Tenant Boundary Example

Tenant A Group:

```
environment:dt.security_context IN ("tenant-a");
storage:dt.security_context IN ("tenant-a");
settings:dt.security_context IN ("tenant-a");
```

Tenant B Group:

```
environment:dt.security_context IN ("tenant-b");
storage:dt.security_context IN ("tenant-b");
settings:dt.security_context IN ("tenant-b");
```

Platform Team (cross-tenant):

```
environment:dt.security_context IN ("platform", "shared");
storage:dt.security_context IN ("platform", "shared");
settings:dt.security_context IN ("platform", "shared");
```

Entity Enrichment for Multi-Tenancy

Use Entity Enrichment rules (Settings > Entity Enrichment) to assign tenant context:

```
# Entity Enrichment rule based on host group
Rule: Assign security context from host group
Entity Type: Host
Condition: Host group name contains "tenant-"
Action: Set dt.security_context = {hostGroup.name}
```

Alternatively, use **host properties** to set context at deployment:

```
# Set via OneAgent installer
--set-host-property=dt.security_context=tenant-a
```

7. Boundary Testing

Verify boundaries work as expected before production use.

```
// Check security context distribution on services
fetch dt.entity.service
| summarize count(), by:{dt.security_context}
| sort count() desc
| limit 20
```

```
// Find entities without security context (boundary gaps)
fetch dt.entity.service
| filter isNull(dt.security_context)
| fields entity.name, tags
| sort entity.name
| limit 50
```

```
// Check host security context coverage
fetch dt.entity.host
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context)),
    missing = countIf(isNull(dt.security_context))
| fieldsAdd coveragePercent = round(100.0 * withContext / total, decimals: 2)
```

```
// Verify logs have security context
fetch logs, from: now() - 1h
| summarize
    total = count(),
    withContext = countIf(isNotNull(dt.security_context))
| fieldsAdd coveragePercent = round(100.0 * withContext / total, decimals: 2)
```

Testing Methodology

1. **Create test group** with the boundary

2. **Add test user** to the group
3. **Log in as test user** (or impersonate)
4. **Verify visible entities** match expected
5. **Verify hidden entities** are not visible
6. **Query data** to confirm filtering works

Validation Checklist

Test	Expected
List services	Only boundary-included services
Query logs	Only logs with matching context
View settings	Only settings with matching context
Access denied entity	Empty result or error

Next Steps

With boundaries configured, complete your IAM implementation:

Recommended Path

1. **IAMADM-06: User Lifecycle and Provisioning** - Automate user management
2. **IAMADM-07: Audit Logging and Compliance** - Monitor access patterns
3. **IAMADM-08: Multi-Environment IAM** - Scale across environments

Boundary Checklist

Before moving on, ensure you have:

- ☐ Understood the three-domain model
- ☐ Designed your security context strategy
- ☐ Created boundaries for each team/group
- ☐ Configured entity enrichment for context assignment
- ☐ Tested boundaries with real users
- ☐ Verified no entities are missing context

Summary

In this notebook, you learned:

- Boundary fundamentals and how they differ from policies
 - The three-domain model (environment, storage, settings)
 - Boundary syntax and operators
 - Security context strategy and naming
 - Five common boundary patterns
 - Multi-tenant isolation design
 - Boundary testing and validation
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References

- [Permission Boundaries](#)
 - [Security Context](#)
 - [Entity Enrichment](#)
 - [Host Properties](#)
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